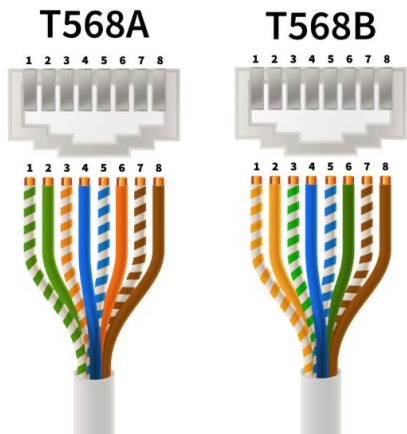


1. RJ-45 Connector Pin Layout (T568A and T568B)



Pin	T568A Color	T568B Color
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

2. Comparison of Ethernet Standards

Ethernet Standard	Speed	Cable Type	Maximum Cable Length	Connector Type
10BASE-T	10 Mbps	Cat3, Cat5, Cat5e	100 meters	RJ-45
100BASE-TX	100 Mbps	Cat5 or Cat5e	100 meters	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e or Cat6	100 meters	RJ-45

3. Connecting Two Laptops Directly Using Ethernet

To connect two laptops directly, a crossover Ethernet cable is traditionally used.

Reason:

- A crossover cable swaps the transmit (TX) and receive (RX) wire pairs, allowing the laptops to communicate directly without a switch or hub.
- However, most modern laptops support Auto MDI-X, which automatically detects the cable type. Therefore, a straight-through cable also works on most modern devices.

Answer for assignment:

A crossover cable is used because it crosses the transmit and receive pairs, enabling direct communication between two similar devices such as laptops. Modern laptops with Auto MDI-X can also work with a straight-through cable.

4. Real-World Devices That Use Ethernet

Device	Location	Likely Ethernet Standard
Desktop Computer	Home/College Lab	1000BASE-T (1 Gbps)
Wi-Fi Router	Home	1000BASE-T (1 Gbps)
Network Printer	College Office/Home	100BASE-TX (100 Mbps)